Business Management for IT (IE3060)
3rd Year, 1st Semester

Report

AI-Driven Transparent Food Review System for Uber Eats

Submitted to

Sri Lanka Institute of Information Technology

In partial fulfillment of the requirements for the

Bachelor of Science Special Honors Degree in Information Technology

17/10/2025

Declaration

We hereby declare that this project report titled "AI-Driven Transparent Food Review System for Uber Eats" is our own original work conducted under the supervision of Lecturer: Ms. Colinie Wickramaarachchi

All sources consulted and referenced have been duly acknowledged.

We fully understand the consequences of academic dishonesty and certify that this work complies with the university's ethical guidelines.

Submitted by: Group 1

Table 0-1: Student details

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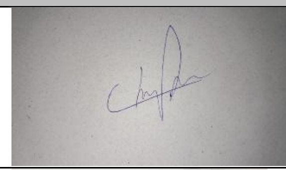
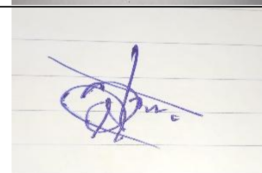
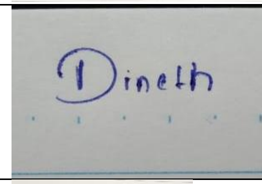
IT23184176	C P Wanniarachchi	
IT23199194	K P D M Somachandra	
IT23150652	D C B V D S Wimalasena	
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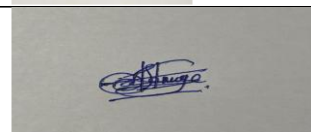
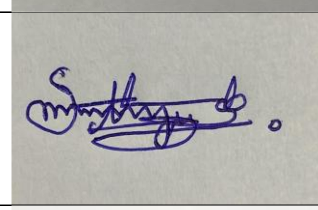
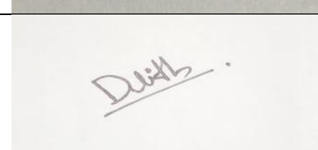
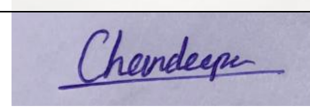
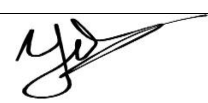
IT23363052	K H S Damsiluni	
IT21258176	W W S M Fernando	
IT23409828	Jayasekara D T S	
IT23362994	H M C M B Herath	
IT20054854	Y K Wijerathna	
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Abstract

This report presents a well-defined technical, marketing, management, and financial analysis of an AI-Driven Transparent Food Review System for Uber Eats. It has been designed to enhance customer trust and transparency within the review platform. This report introduces a structured, AI-powered review system that automatically categorizes customer reviews into key aspects such as food quality, packaging, portion size, etc. This innovation helps new customers to make well-informed decisions. Also, it increases the restaurant's accountability and eventually the Uber Eats brand image.

This report evaluates several dimensions from a management perspective. It defines how the technical implementation should be carried out with a customer-driven marketing strategy.

It proposes frameworks such as Kotter's 8-step model and the Rational Decision-Making model to ensure smooth implementation. The financial and accounting justification examines the technical budget, marketing costs, and return on investment, which demonstrate the financial and positive ROI capability of the initiative. This report analyzes Sri Lanka's inflation trends and customer price sensitivity from an economic perspective. It shows how the transparent review system can establish customer loyalty and market stability for Uber Eats

Ultimately, by integrating technical implementation, management strategies, financial analysis, and economic justifications, this report demonstrates how Uber Eats can use AI technology to build customer confidence, establish customer transparency, and achieve a long-term competitive advantage as a food delivery giant in the global market.

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Chapter 1 – Management Foundation

1.1 Introduction to Uber Eats

Uber Eats is a leading food ordering and delivery platform that provides its services online to both customers and end users via a mobile application. Uber Eats is currently operating in 30 countries around the world, including Sri Lanka. It is a successful and very popular food delivery platform centered on connecting customers with merchants(restaurants) via delivery partners. Uber Eats app connects the customers with a range of local restaurants that are registered on Uber, where the customers can order from the menus offered by the restaurants (Uber Eats Help Center, 2024). The core part of this Uber Eats is the mobile app, which acts as the interface that connects customers with restaurants and delivery partners. Uber Eats highly relies on technology, such as data-driven decisions, real-time operations, and availability for the end users of the app. Through continuous innovations and customer-friendly approaches, Uber Eats has become very successful in the online food delivery market while competing with other competitors. Uber Eats has a successful delivery operation where the ordered food is delivered within a few minutes at the customer's doorstep, which makes it one of the most competitive and customer-centric brands in the global online food delivery industry.

1.2 Link to Marketing Problem

1.2.1 Problem Definition

Although Uber Eats has become a leading giant in the food delivery system, the biggest challenge it faces is its lack of transparency in the existing review system. That review system has led to a reduction in the trust and confidence of the customers towards Uber Eats and has finally damaged the brand of Uber Eats.

Currently, Uber Eats has a thumbs up/down and a star rating system, where the end users can rate their experience compared to a given range of stars (Uber Eats Help Center, 2024) (Uber Eats Help Center, 2024). which fails to reflect the real experience in a more informative way. It should include parameters like the food quality, portion, taste, delivery time, price, etc. This gap reduces the confidence of customers in trying out a new restaurant, a new food item which he has never experienced before. This issue leads to dissatisfaction among the customers after making the purchase and consuming it, and then diverting both existing and new customers from the restaurants and eventually from Uber Eats.

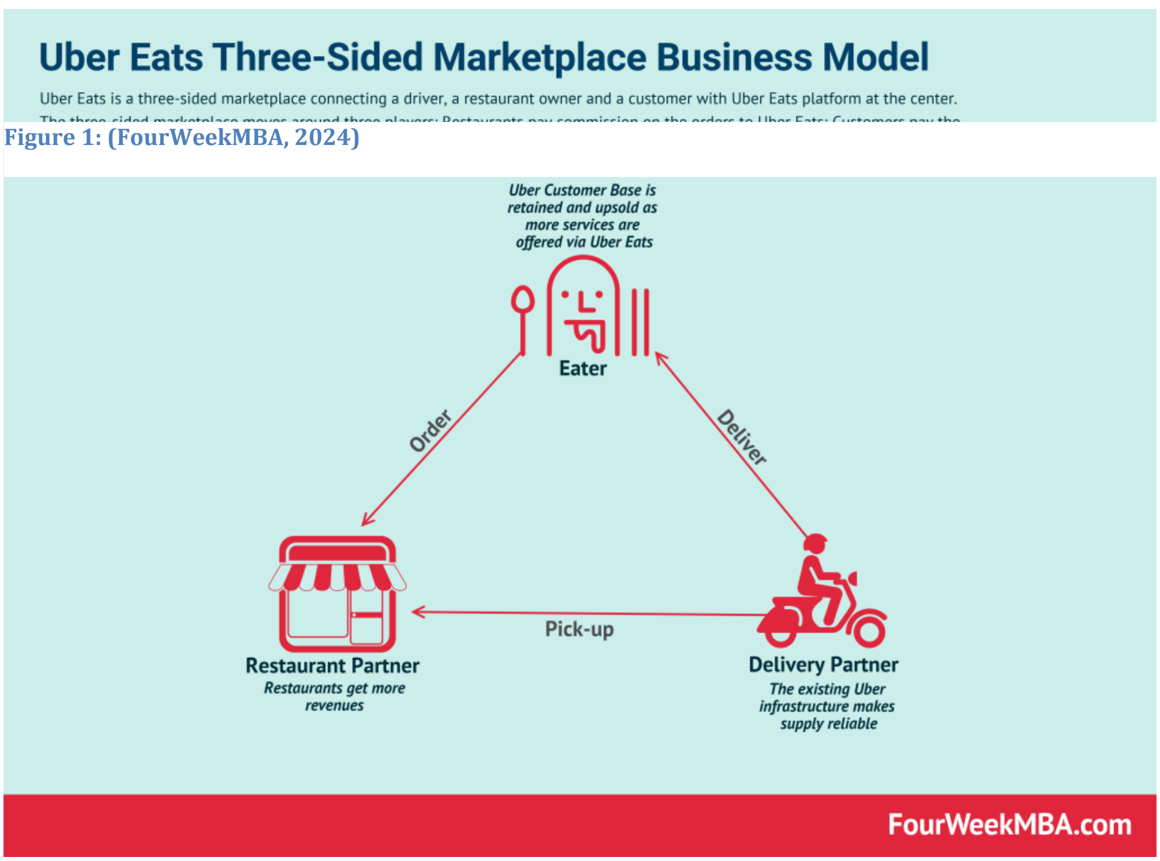


Figure 1: (FourWeekMBA, 2024)

From a marketing perspective, this lack of transparency leads to the following issues,

- Brand image: Customers tend to idolize Uber Eats as a primary food delivery platform and rate it down.
- Customer retention: Decreasing reordering by dissatisfied customers.
- Promotional failures: Customers start to question the experience they are getting through the platform
- Merchant trust: Restaurants start to lose customers and miss the new customer opportunities due to a lack of a proper review system. They miss the chance to get verified feedback from the customers.

1.2.2 Proposed Solution

To address those marketing challenges, Uber Eats needs to categorize its solutional approach into two main parts: Technical Integration & Marketing Integration.

- Technical Integration: AI-driven food review System enables the reviewer to type their review in simple text and add a picture of the food that they got delivered. Then the AI review system analyzes the textual customer feedback, identifies, and places the review under the identified review category of the food item. So, this approach leads to a more informative review, even making the customer review placement task very simple. Those reviews will be added near each food item to make each review unique and item-specific. It will enhance transparency by allowing other customers to be informed and pre-notified about the food items that they are going to purchase. Additionally, customers can rate the food item via a star-based review system. Finally, the total star ratings for a food item will be considered for a display of the average rating number near each food item, and then the overall ratings of all food items to rate a particular restaurant. This system will increase the demand for the restaurants that sell food of better quality without fooling customers. It will increase the trustworthiness of the service provided by both restaurants and Uber Eats, which will eventually attract both new customers and restaurants towards Uber Eats than the other competitors. Appendix:

Table 2: Comparison of issues and solutions

Current System Issue	Proposed AI-Based Solution
Limited to stars and thumbs up	Text reviews are analyzed and categorized by AI
Reviews lack context	Categorized reviews (eg, portion, weight)
Hard for new customers to judge	Filtered more informative reviews with pictures
Manual moderation	AI automation to ensure efficiency and fairness.

- Marketing Integration: This phase aims to turn the new technical integration into a Brand advantage. A new marketing program can be carried out to promote this new system as a customer trust-building, innovation. In this promotion, Uber Eats can encourage customers to “see before buy”. The promotional campaign can be carried out with these elements.
    1. Core Message: “Transparency Builds Trust”.
    2. In-App promotion: Highlight real reviews and photos.
    3. Social Media Campaigns: #transparencyBuildsTrust, small video clips about customers’ opinions on this initiative, customer review stories, etc.
    4. Reward systems: A reward system featuring ‘1 of Top 10 Transparent Restaurants” and “Verified” badge as a co-branding strategy.
    5. Customer awareness: Short demonstration clips and banners to educate customers about ‘how to use this new review system’, and
    6. Branding with AI: Highlighting the integration of AI to attract the young generation to try out the feature.

  It can be carried out as a marketing strategy to reinforce Uber Eats’ lost customer base with the promise of convenience, transparency, and user-friendliness.

1.3 Management Challenges

Implementing the marketing program for the AI-based transparent review system introduces several internal management challenges for the company itself. The marketing initiative introduces a new challenge, which is to get the acceptance of the main stakeholders: Customers, Restaurants, and Delivery partners.

To manage this transformation effectively, the company must focus on the following areas which are leadership, organizational structure, decision-making processes, change and transition management, and risk management.

1.3.1 Leadership

Strategic, resilient, and adaptive leadership is expected with the capabilities of aligning AI technology with brand communication. The success of the marketing program heavily depends on,

- Vision alignment: Leaders should initiate a customer-friendly marketing message that gives a clear idea of the technical function of the AI-based review system to rebrand Uber Eats.
- Cross-Functional Coordination: Marketing heads must ensure smooth collaboration between product managers, UI/UX teams, data scientists, and local marketing offices.
- Stakeholder Communication: Leadership must emphasize the competitive advantage, long-term marketing benefits, and openly communicate with them to avoid resistance from restaurants and delivery partners.
- Adaptive Leadership: A transformational leadership with tech tech-friendly mindset will help to align different teams to be innovative, adaptive, risk-taking, creative, and strategic during the marketing campaign execution.

1.3.2 Organizational Structure

Uber Eats is a global giant. To launch a marketing campaign across 30 countries around the world, that organization needs to be top-to-bottom well-structured and compatible with any political, economic, and cultural setting.

Main challenges of the marketing program include:

- Coordination across regions: Since Uber Eats operates in 30+ countries with different political, cultural, and economic settings.
- Resource allocation: Allocating budgets, staff, and tools across regions to maintain continuous marketing operations.
- Integration with other departments: The Marketing program needs the collaboration of all development, Finance, HR, product management, and legal departments to ensure the marketing adheres to all rules. regulations and technical guidelines.
- Feedback Integration: Weekly internal meetings to discuss and analyze campaign results, find new metrics, and address new challenges based on customer feedback.

1.3.3 Decision making

Decisions are the main thing that drives the marketing program to achieve the expected results. Challenges in decision making include,

- Cultural, political, and regional specific backslashes: Managers should be careful about the marketing approaches they are going to carry out based on different environmental settings. They should make their decisions by respecting every party involved. Top-level managers can discuss with the local managers to get a regional-specific decision to carry out the marketing program.
- Large data set to make decisions on: The Decision process must be capable of interpreting and analyzing the large result set generated from the marketing campaign, which may include customer feedback, marketing successes, etc.
- Prioritization conflicts: A discussion between the marketing teams and the data scientists should be carried out to decide how much weight should be applied to each stakeholder (customer, restaurant, delivery partner) during the marketing process.

Rational Decision-Making Model (Creately, 2024).

- Step 1: Problem Identification: Reduction of repeat orders due to unsatisfied customers because of low transparency in the existing review system. It leads to loss of loyal customers and an inability to attract new customers.
- Step 2: Information Gathering: Marketing and data science teams collect data from customer feedback, social media analysis, and marketing campaigns held to assess the success of marketing the new AI review system.
- Step 3: Develop alternatives: Uber Eats evaluates marketing alternatives.
    - Simplify the new review system explanation process, targeting users who lack much technical knowledge.
    - Highlighting the user friendliness of the new system by demonstrating the review operation.
    - Emphasizing the drawbacks of the existing review system of other food delivery platforms to get a competitive advantage.
    - Showcasing how the system has succeeded in other regions.
- Step 4: Evaluate Alternatives
    - Each alternative can be assessed based on increased customer base, newly entered review counts, comparing the marketing results by region, cost of the marketing process, and response from the other competitors.
- Step 5: Select the best alternative
    - Uber Eats can select the best marketing approach based on the evaluation done in step 4. The marketing approach will differ from region to region.
- Step 6: Implementation of the Decision

    - This step is discussed in detail in the "1.4 Implementation Plan".
- Step 7: Evaluate the Decision
    - After the implementation of the Decision (step 6), Uber Eats can analyze KPIs such as new customer adoption rate, repeat order ratios, patterns of the review counts, social media feedback, and count of new Uber Eats account creations.

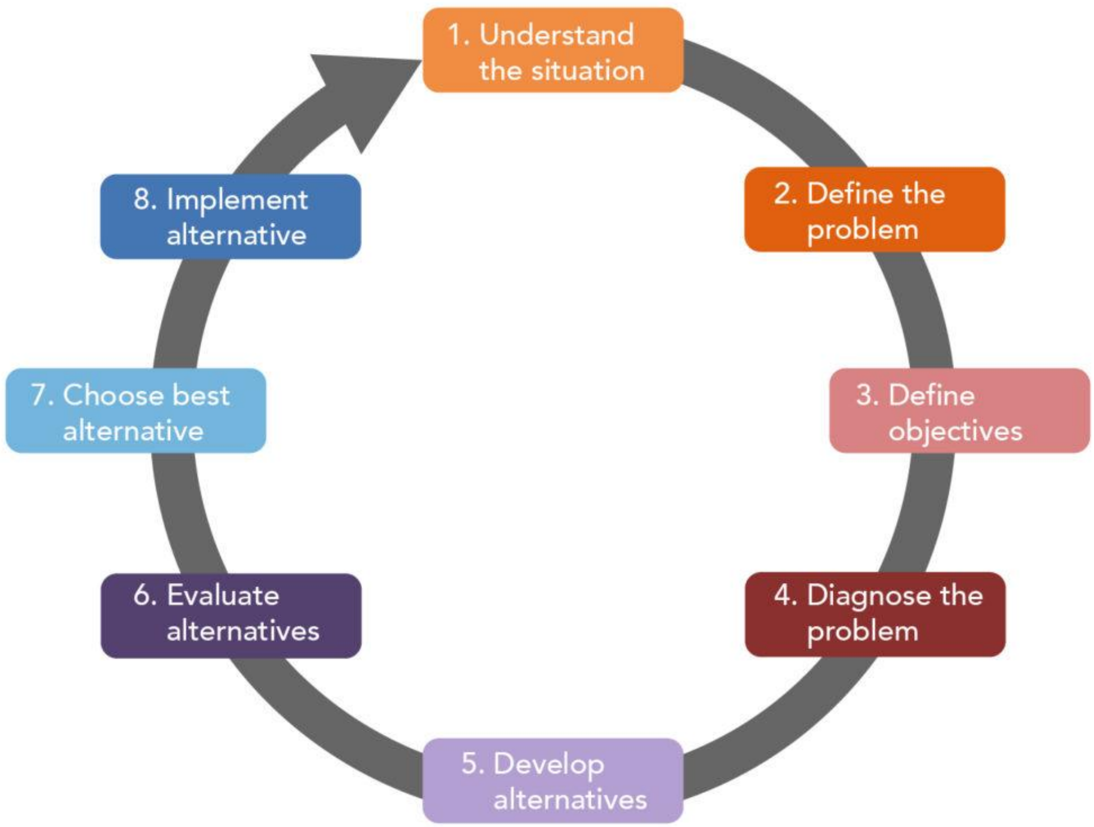


Figure 2: (Lumen Learning, 2024)

1.3.4 Transition Management

This is the most critical part, which introduces both internal and external challenges. Challenges may include,

- Restaurant's resistance:
    - Restaurants may resist this new initiative due to the public visibility of bad reviews. Uber Eats should show the new opportunity to grow as a quality restaurant and to increase visibility through verified reviews.
- Delivery Partner resistance:
    - Showing the new opportunity to be verified through the platform. Creating a reward system to celebrate their successes.
- Customer resistance:
    - Customers with low technical literacy may avoid using this new AI-based review system. A marketing campaign driven by educational content, sample demonstration videos, and in-app prompts can showcase the benefits of this new feature. Statistical predictions can be used to emphasize how the placement of a review can give an advantage in return.

Uber Eats can follow 'Kotter's 8-Step Change Model' for a smooth transition (Whatfix, 2024).

Especially the 1st step of the model, which is,

Create Urgency: Highlighting the drawbacks of declining honest customer reviews and applying competitive pressure.

These approaches ensure stakeholder acceptance, and they can be enhanced via feedback loops. Stakeholder training sessions can be carried out to make them more familiar with this system and to make the marketing program successful.

- Internal Alignment and campaign acceptance: Marketing teams must be trained to simply communicate the technical features of the new system to avoid technical jargon. They collaborate with technical team members who have some marketing knowledge for this campaign.

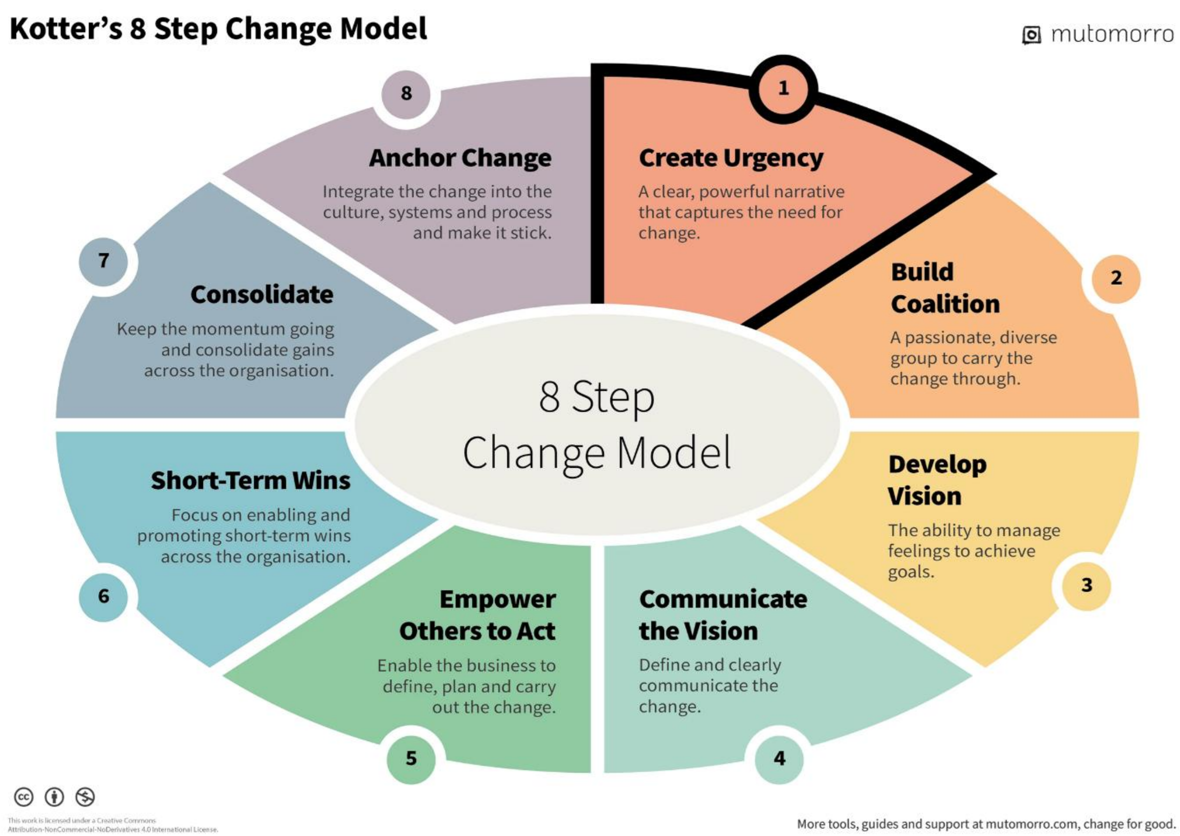


Figure 3: (Mutomorro, 2024)

1.3.5 Risk Management

Several cultural, ethical, brand, and operational risks may arise since this marketing program is about a highly technical feature.

Table 3: Risk and Mitigation strategies

Risk Type	Challenges in the Marketing Program	Mitigation strategy
Negative reaction from Restaurants	Restaurants publicly opposing the new feature	Reaching each restaurant partner and joining them to the marketing program, which eventually gives them a branding opportunity.
Data privacy concerns customers.	Customers are doubting how AI interprets their personal data	Getting certified in preserving data privacy from local government bodies and reputable international associations, such as the IEEE.
Message Dilution	Inconsistent communication across regions.	Keeping centralized marketing guidelines and brand tone-of-voice manuals.

1.4 Implementation Plan

Uber Eats can adopt an implementation plan for this marketing program that involves several phases, spanning 12 months.

Phase 1: Pilot marketing campaigns (months 1-5)

- Market Selection: Choose 2-3 major urban cities with different demographics. Regions with high app engagement and strong restaurant partnerships that were loyal to Uber Eats for years.
- Target Audience: Early tech adopters and frequent buyers who agree to feature the transparent review pilot.
- Marketing activities: Launch in-app promotional banners introducing new features, social media awareness campaigns, collaboration with food influencers and bloggers.
- Result evaluation: Measuring adoption rate from new review counts, tracking customer satisfaction scores, brand sentiment analysis, and analyzing the average ranking behaviors of both food items and restaurants.

Phase 2: Evaluation and optimization (months 4-6)

- Data analytics: Evaluating customer engagement metrics (review submission rate, time spent on review pages). Using AI-driven marketing analytics to segment users who respond positively to transparency messaging.
- Marketing strategy refinement. Adjust message tone and visuals to align with cultural and regional audiences. Conduct webinars and training for restaurant owners to use Uber Eats Manager to manage reviews effectively.

Phase 3: Full-scale rollout [months 7-12]

- Global Launch Campaign: Officially announce the AI-Driven Transparent Review System across all major regions.
- Create localized Marketing content for different regions, aligning with language, culture, and consumer behavior.
- Collaborate with global restaurant chains to co-brand and promote the feature.
- Partnering with digital payment services like Visa and PayPal to offer small cashback rewards for a selected number of first-time reviewers.
- Introducing badges for frequent reviewers. E.g., “Trusted Voice”, “Taste Explorer”.
- Monthly KPI reviews can be carried out, including,
    - Customer adoption rate, review viewing level, Customer reorder and retention rate behavior, Calculating revenue behavior from marketing campaigns.

Phase 4: Post-implementation review.

- Compare real vs projected ROI, customer satisfaction rate, and revenue increases.
- Retraining the AI model periodically with new review data to improve categorization accuracy.
- Keep integrating customer feedback loops in app notifications and social media.
- Publish success stories of restaurants and delivery partners across the globe.

Final Expected outcome:

- A sustainable marketing ecosystem and a transparent review system that ensure Uber Eats remains a market leader in customer trust-based digital delivery experience.

Chapter 2 – Accounting & Financial Justification.

2.1 Projected Financial Outcome (Budget)

2.1.1 Software Development & Integration

Component	**Description**	**Estimated Cost**
Backend Integration	New APIs to connect Uber Eats' existing data with the new AI module	$8,000
Database Expansion	Adding new tables for reviews, food dimensions, and AI outputs into the existing database.	$3,000
Frontend Integration	Updating the UI for displaying categorized reviews, price charts, labeled restaurants, and portion visuals.	$7,000
User Feedback Update	Modify the existing customer review submission to allow photo uploads and food dimensions feedback.	$2,000
Testing & Debugging	Testing the new module for bugs and vulnerabilities, and compatibility between different Operating Systems	$2,000

Subtotal = $22,000

2.1.2 AI model development

Component	**Description**	**Estimated Cost**
NLP Classifier	Classify a user review into pre-defined categories	$5,000

Sentiment Analyzer	Process the user review and determine customers' emotional tone: positive, neutral, or negative.	$2,000
Image Processor	Recognizing food products and their dimensions (size, texture, color etc.) from uploaded photos to corresponding categories.	$7,000
Review Aggregator	Collect all user reviews, analyze, organize and summarize them to provide an overall opinion for a specific product from a company.	$2,000
Reputation Scorer	Calculates numerical or categorical scores to represent the overall reputation of the product from the restaurant based on multiple data points.	$3,000

Subtotal = $19,000

2.1.3 Analytics and data visualization

Component	Description	Estimated Cost
Price History Tracking Engine	Subsystem that records, monitors, and analyzes product price changes over time.	$4,000
Restaurant Dashboard	An interface for restaurants to view items' reputation, price trends, and review summaries.	$5,000
Customer View Integration	Modify customer view by adding visual charts and filters.	$3,000

Subtotal = $12,000

2.1.4 Cloud Infrastructure Deployment

Component	Description	Estimated Cost
Cloud Hosting (AWS, Microsoft Azure)	Integration of the new AI model into the existing cloud platform. Includes database and analytics.	$1,000/month × 6 = $6,000
Model Hosting & API Management	Deploying all through AI models so that the application can easily access.	$3,000
Cloud storage	Need additional cloud storage to store user added photos of products.	$1,000

Subtotal = $10,000

2.1.5 Project Management & Maintenance

Component	Description	Estimated Cost
Project Management	Feasibility study, requirement analysis, design, and monitoring of the project progress.	$3000
System Maintenance	Bug fixes, regular backup & recovery, updates	$4000

Subtotal = $7,000

2.1.6 Total Estimated Budget for Technical Implantation

Category	Estimated Cost (USD)
Software development & Integration	$22,000
AI model development	$19,000
Analytics & data visualization	$12,000
Cloud Infrastructure Deployment	$10,000
Project Management & Maintenance	$7,000

Total Cost = $70,000

2.1.7 Marketing Campaign Budget Breakdown

Component	Description	Estimated Cost
Digital Advertising	Paid promotions on social media platforms across targeted urban markets	$4000
Creative Design & Video Production	Short social media videos, app demos, and customer education content.	$2000
In-App promotions & Discounts	Rewards for top reviewers and loyalty discounts	$3000
Market Research & Analytics	Real-time tracking of campaign KPIs, customer response rates, and adoption behavior analytics.	$1500
Influencers	Collaboration with regional food influencers, bloggers, and content creators.	$2000
Public Relations	Restaurant partner events and cross-brand campaigns	$2000
Training & Internal Communication	Training restaurant partners, delivery partners, internal teams, and customers.	$1000

Subtotal: $15500 (for 12 months of campaign)

Total Estimated Budget for Technical Implementation & Marketing Campaign: $85,500

2.2 Projected Revenue Impact.

According to *Business of Apps* (2025), Uber Eats has experienced rapid year-on-year revenue growth since 2020, reaching a revenue of $13.7 billion last year. The graph below shows the revenue increase in visual detail.

The marketing program aims to generate revenue growth through enhanced customer trust and engagement, leading to higher order frequency and lower refund rates.

Uber Eats revenue 2017 to 2024 ($bn)

Year	Revenue ($bn)
2017	0.6
2018	1.5
2019	1.9
2020	4.8
2021	8.3
2022	10.9
2023	12.1
2024	13.7

Figure 4: (Business of Apps, 2025)

Uber Eats gross bookings 2017 to 2024 ($bn)

Year	Gross bookings ($bn)
2017	3.1
2018	7.9
2019	14.5
2020	30.2
2021	51.6
2022	55.7
2023	67.8
2024	74.6

Figure 5: (Business of Apps, 2025)

Based on pilot assumptions and market research:

Increase in order volume: +5% (due to improved brand perception and transparency).
Increase in repeat purchases: +8% (customer loyalty and confidence).
Reduction in refund/complaint costs: –15% (due to clearer customer expectations).

Expected ROI period: 9–12 months post-launch.

Since the revenue in 2024 was $13.7 billion, when we integrate the AI-driven review and reputation management system, it is estimated that overall revenue could increase by **5%** in the following year, reaching approximately **USD 14.385 billion.** This is due to factors such as increased user trust, transparency, and engagement. Since the order volume is expected to increase by 5%, we are expecting a 5% increase in Uber Eats' gross bookings.

2.3 Return on Investment Analysis

Increase of revenue due to AI model = $13.7 x 5% billion = 0.685 billion (Average)

ROI margin = 0.685 billion / 13.7 billion = 685 million / 70 thousand = 9785.7

So, compared to the deployment cost of $70,000, the increase in revenue is 9 times higher. So, this shows strong financial feasibility.

Current gross bookings = $74.6 Billion

Increase in gross bookings = $74.6 Billion * 5% = $3.73 Billion additional bookings

2.4 Cost Control Recommendations.

1. Deploy lightweight AI models instead of large language models. (DistilBERT or MiniLM) (Zilliz, 2024) (DeepWiki, 2024).
2. Optimize data storage by archiving historical data: data that is not accessed frequently in cheaper cold storage. The best option is to use Google Cloud Coldline storage if that historical data is accessed at most once in a quarter (3 months) (TechTarget, 2024).
3. Monitor usage with budgets and alerts
4. Use pre-trained NLP models to save the time needed to train the AI model to process business data. Therefore, the final product can be deployed and integrated into Uber Eats much faster. Thus, it reduces the time taken for deployment.
5. Modular microservice integration. Develop this AI module as a separate microservice without rebuilding the core system.
6. Keep summarized analytics instead of full raw data when possible. This reduces the database size.
7. Use open-source frameworks such as PyTorch and HuggingFace instead of pain Machine Learning services.

2.5 Cost control & Investment Recommendations for Marketing program

- Leverage Data-Driven Advertising: Use AI to make predictive analytics to optimize the marketing campaign and make future forecasts.
- Partnering with influences based on performance: Pay and recruit based on engagement or conversation rather than social hype.
- Optimize Cross-Channel Marketing: Integrate in-app, email, and social media ads to reduce redundancy.
- Adaptive region-wise budgeting: Allocating more budget to high-performance regions while maintaining a low range for testing new markets.
- Monitor KPIs continuously: Use the dashboard to track cost-based analytics in real time.

2.6 Link to Poster

As shown in the competitive advantage section of the poster, our new AI-based solution will give Uber Eats a competitive advantage among food delivery services throughout the world.

"In its home market, Uber has been in fierce competition with DoorDash, which surpassed Grubhub to become the most popular food delivery service in 2019. Uber bolstered its position in July 2020 with the acquisition of Postmates billion. Uber takes a cut of between 20 to 30 percent of all orders, a high margin but one that has become commonplace with the growth of DoorDash. In some regions, Uber Eats is profitable, but the company as a whole is bolstered by its profitable mobility segment." (Business of Apps, 2024)

According to the above statement and by observing the following graph we obtained from the website, Uber Eats has a very high competition with DoorDash in the United States. We believe that Uber Eats might have an exponential increase in revenues and bookings at least by 2030, potentially surpassing that of DoorDash.

Chapter 3 – Economic Environment Analysis

3.1 Selected Economic Factor: Inflation & Food Price Sensitivity

Sri Lanka's economy has faced severe inflation over the past few years. In 2022, headline inflation peaked at 49.7%, while food inflation hit an all-time high of 94.9%. Though the overall inflation rate fell to 1.5% as of September 2025, food prices remain very high. The cost of living is still a significant issue for many Sri Lankans. According to the Department of Census and Statistics, food and non-alcoholic beverages make up over 40% of the average household’s monthly expenses (Trading Economics, 2024) (Department of Census and Statistics, Sri Lanka, 2024). This ongoing economic pressure has made consumers, especially those in urban areas, very sensitive to price changes and highly focused on getting good value when they shop (Trading Economics, 2024) (Department of Census and Statistics, Sri Lanka, 2024) (Central Bank of Sri Lanka, 2024).

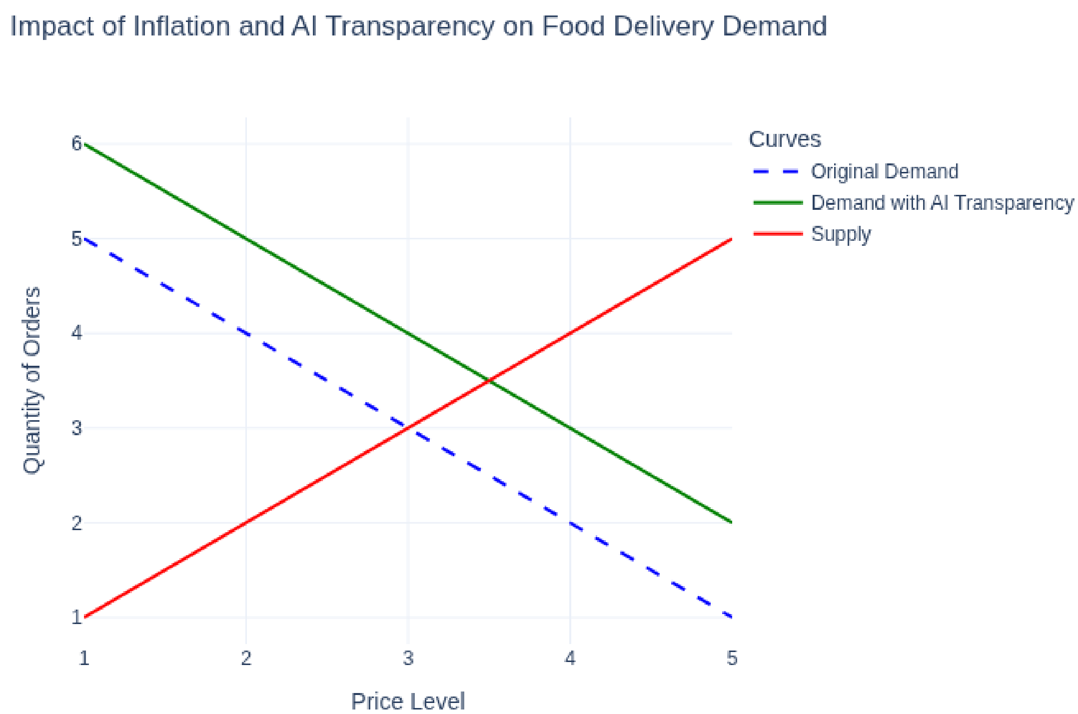
3.2 Impact on Uber Eats & Target Market

Uber Eats' main customers in Sri Lanka are mostly urban youth and professionals. These groups have been hit hard by the country’s economic difficulties. With wages staying the same and living costs going up, consumers are now more careful about their spending, including food delivery. Research shows that urban youth in Sri Lanka are tech-savvy and financially conscious. They often use mobile apps to compare prices and find the best deals before they buy.This means that demand for food delivery services can change quickly based on price or perceived value. For Uber Eats, it is crucial to build customer trust, be transparent, and show clear value to keep and grow its customer base (KDU Journal of Multidisciplinary Studies, 2024).

3.3 Benefits of the AI Review System

The introduction of an AI-driven review system directly responds to current economic realities. By providing honest, categorized reviews that cover portion size, taste, delivery time, and real customer images, the system helps users make informed decisions and lowers the chance of disappointment. This transparency builds trust, which is vital when every rupee matters. Even with high prices, customers are more likely to keep using Uber Eats if they feel confident about the value and quality of their orders. The AI review system also motivates restaurants to uphold high standards, knowing their offerings will face clear, public feedback. In this way, the system helps stabilize demand and supports customer loyalty, even in a tough economic environment.

3.4 Graph/Model Explanation



Graph: Impact of Inflation and AI Transparency on Demand for Food Delivery in Sri Lanka

The demand and supply graph shows how inflation and AI transparency affect food delivery demand in Sri Lanka. The blue dashed line represents the original demand curve before inflation. Inflation pushes this curve to the left, meaning there is lower demand at higher prices. However, the introduction of AI-powered transparency shifts the demand curve back to the right, indicated by the green line, partially balancing the decline. The red line represents the supply curve, which is assumed to be constant. This model illustrates how transparency and trust can help keep order volumes stable despite ongoing price pressures.

3.5 Tie to Customer Persona & Targeting

This economic logic is directly linked to the customer personas and targeting strategy highlighted in the poster. In offering solutions to Sri Lankan urban youth and professionals' pain points of distrust, fear of quality problems, and need for transparency, the AI review system keeps Uber Eats remain significant, competitive, and robust even during a change in economic times. This alignment underscores the overall customer-focused marketing strategy.

Chapter 4 –Integration & Conclusion

4.1 Integration of AI-Driven review & rating Module into Uber Eats

This section explains how the proposed AI-based review categorization and rating system can be integrated into a mature food delivery system like Uber Eats. The goal is to show how data flows, what components are updated or modified, and how the system remains scalable and maintainable.

By connecting real-time customer reviews, categorized sentiments, and star ratings to Uber Eats' existing analytics infrastructure, this integration enables data-driven marketing, targeted promotions, and customer trust-building through transparent, organized reviews.

4.1.1 High-level overview of proposed upgrade

- Accept user inputs submitted as text reviews, photos, and star ratings.
- Use an AI engine to classify each review into categories. (Portion, Taste, Packaging, Delivery time).
- Store categorized feedback.
- Compute and update average ratings per food item. From that data, we will be able to calculate an overall rating for the restaurants.
- Expose categorized reviews to customers. (Can be filtered by category).
- Ensure compatibility with existing systems such as order management, menus, merchant dashboards, and customer app.

At a high level, this proposed AI engine acts as a pipeline between the customer app and the analytics modules.

Component	Integration	Allocated task
Customer app (IOS / Android)	Add "submit review"	After an order is delivered, the app shows a screen to enter text review, photo uploader, and star rating
API gateway (Frontend, Backend)	Review endpoints	Add new endpoints to get reviews, get item ratings, etc.
AI review module	Receive review payload, runs classification	Performs image processing
Application layer Backend	Orchestration	Check and insert new categories if needed.
Database	Schema extensions	Add tables to store categories.
Merchant dashboard	New UI	Merchants can see new options and use them.
Caching / Analytics pipelines	Periodic recalculation	Keep aggregated metrics

4.1.2 Real-time performance and scalability considerations.

Uber Eats handles a big data traffic at once. Because of that, this integration must respect performance, scalability, and reliability.

1) Microservice architecture
   The AI review system should be its own service, communicating via APIs to decouple it from the main application structure.
2) API gateway
   All requests can be funneled through this, which routes to internal microservices such as reviews, orders.
3) Asynchronous processing
   AI classification might be done asynchronously. So, the reviews can be accepted immediately.
4) Real-time data pipeline and analytics
   Uber has a good pipeline infrastructure already. This new integration will not affect that.
5) Data privacy and moderation
   Implement moderation logic in the review system (Filtering offensive content)

4.1.3 Challenges and risk mitigation

1) Misclassification – Some reviews may address multiple aspects. (Small portion but tastes good.)
2) Scalability under high volume – Optimize model interface and use GPU/CPU scaling if needed.
3) Image storage – There should be a suitable storage space just for image reviews.
4) Data consistency – Race conditions when two reviews are inserted same time for a single item (use atomic updates as a solution).
5) User privacy – Ensure images by content policies
6) Merchant manipulation or abuse – protect merchants from unfair reviews.

4.1.4 Marketing impact of integration

1) Data-driven campaigns: Marketing teams can use categorized insights to run precise campaigns.
2) Improved conversion: Transparent, category-based reviews can increase the trust of new users.
3) Restaurants of the best quality will be promoted automatically.
4) Most importantly, this new integration will position Uber Eats as the most feedback-related and customer-aware platform.

The proposed AI-driven transparent review system and the corresponding marketing program illustrate how management, finance, and economics align to create a customer-driven solution and a business model for Uber Eats.

- From a management perspective, effective leadership, organizational coordination, risk management, and change management practices enable the marketing initiative to effectively translate technological innovation into customer value.
- From a financial perspective, budgets and other economic analytics show both technical implementation and the marketing program's feasibility.
- Through the economic perspective, transparency drives trust and customer satisfaction stabilizes, increases customer demand, and ensures customer market performance.

This report closely aligns with the accompanying marketing poster, which visually integrates these insights.

Management Foundation: The Poster's problem identification and proposed solution sections are underpinned by the management analysis in this report. The application of Kotter's 8-step change model and the Rational Decision-making model emphasizes leadership, decision-making, and stakeholder coordination.

Accounting & financial justification section supports poster's competitive advantage visuals by presenting a detailed budget, ROI calculation, and cost-control strategies.

Economic Environment Analysis complements the poster's Customer Persona & Analysis by linking inflation and demand-supply models to customer behavior.

As a whole, these technical, managerial, financial, and economic sections validate the marketing poster's key message – "Transparency Builds Trust".

4.2 Conclusion

The Advanced AI Review Categorization System represents a step forward in Uber Eats' customer feedback management and utilization system. The incorporation of AI, analytics, and feedback review systems has enabled Uber Eats to convert old and obsolete customer review systems into review analytics, decision systems, and marketing systems.

Uber Eats performs review segmentation into important business elements. Review components such as Portion, Taste, Delivery Time, and Packaging along with pictures. The system has no reviews, helping Uber Eats marketing analytics and merchant reviews systems, analytics, and business growth. It also helps customers experience reviews that are user experience enhanced.

Feedback review systems in marketing, enhanced feedback systems as review segmentation/ classification, help Uber Eats marketing analytics to perform targeted marketing using customer order trends. It helps merchants improve their marketing offers and helps customers improve their confidence in marketing as their purchase decisions are backed with info from reviews and marketing.

By integrating this new system with management leadership, financial stability, and economic adaptability, Uber Eats can reposition itself as the global leader in a trustworthy, AI-enabled food delivery experience.

This initiative ensures that every stakeholder benefits from a more open and mutually rewarding ecosystem. Finally, it concludes: “Transparency builds trust”.

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Appendices

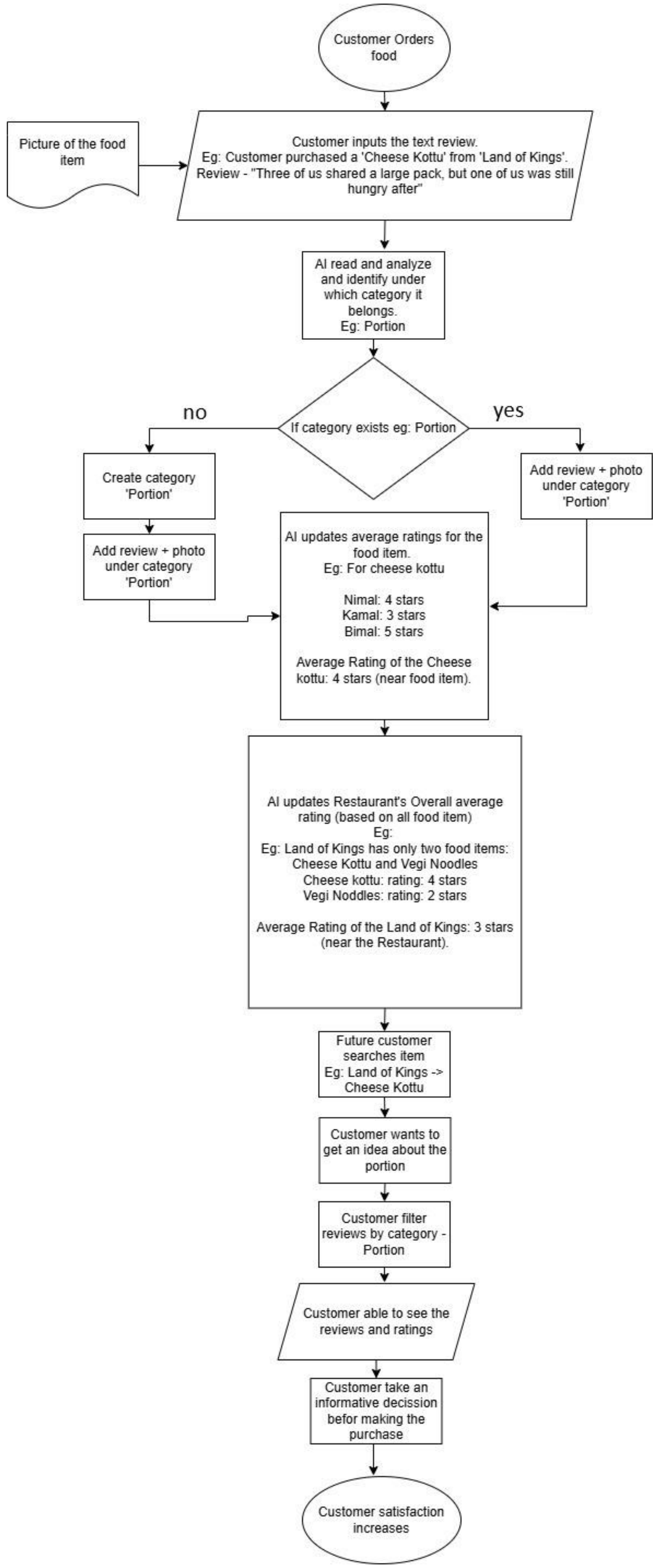

Figure 6: Sample flow chart for customer reviewing process